2.6 Solving Absolute Value Inequalities

Name _____ Period ____ Date _____

Algebra I 2.6 · Student sheet

I can…

- solve absolute value inequalities of the form $|ax + b| < c$.
- solve absolute value inequalities of the form $|ax + b| > c$.
- use absolute value inequalities to solve real-life problems.

Part 1 · Distance thinking

A. Shade all numbers within 2 units of 5:

B. Shade all numbers more than 2 units from 5:

Graph	Absolute value inequality	Compound inequality
A		
B		

Part 2 · Notes: fill in the blanks

$|ax + b| < c$ → _____ $< ax + b <$ _____ (“and”)

$|ax + b| > c$ → $ax + b <$ _____ or $ax + b >$ _____ (“or”)

Isolate the absolute value _____.

Part 3 · Examples (work them with the class)

Example 1

Solve $|x + 4| < 6$ and graph.

Example 2

Solve $|2x - 1| \geq 9$ and graph.

Example 3

Solve $3|x - 2| + 1 \leq 13$.

Example 4

A thermostat is set to 70°F. The temperature stays within 3°F of the setting. Write and solve an absolute value inequality.

Part 4 · Partner practice

Partner 1

Solve $|x - 3| > 5$.

Partner 2

Solve $|x| + 4 < 2$.